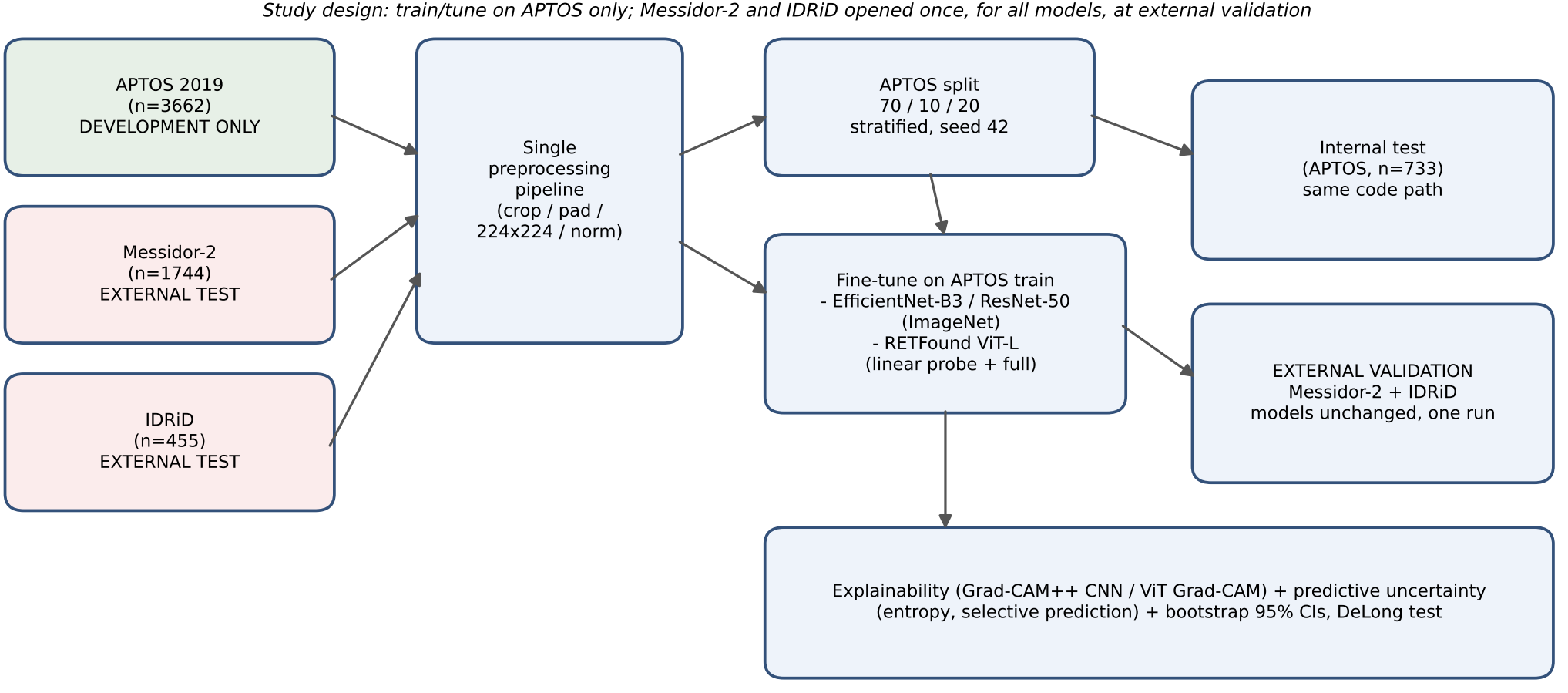
Study design: train/tune on APTOS only; Messidor-2 and IDRiD opened once, for all models, at external validation
APTOS 2019
(n=3662)
DEVELOPMENT ONLY
Messidor-2
(n=1744)
EXTERNAL TEST
IDRiD
(n=455)
EXTERNAL TEST
Single
preprocessing
pipeline
(crop / pad /
224x224 / norm)
APTOS split
70 / 10 / 20
stratified, seed 42
Internal test
(APTOS, n=733)
same code path
Fine-tune on APTOS train
- EfficientNet-B3 / ResNet-50
(ImageNet)
- RETFound ViT-L
(linear probe + full)
EXTERNAL VALIDATION
Messidor-2 + IDRiD
models unchanged, one run
Explainability (Grad-CAM++ CNN / ViT Grad-CAM) + predictive uncertainty
(entropy, selective prediction) + bootstrap 95% CIs, DeLong test